

KKT KOLBE HERMES RGBW Hood

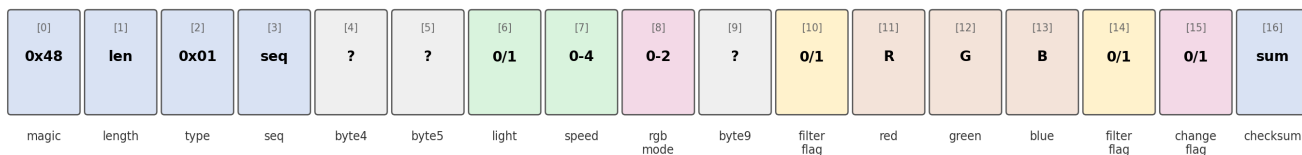
Hekr Protocol Reference — Address Space Map

This document maps every address space in the raw-hex protocol used to control the cooker hood over its Hekr/Wisen cloud connection (hub.hekreu.me:83), reverse-engineered through packet capture, systematic command sweeping, official firmware SDK cross-reference, and community protocol documentation. It covers two distinct address spaces: the **byte layout** within each frame, and the **command ID space** that selects what a frame does.

1. Status Frame Byte Layout

Sent by the device to the cloud (and relayed to Home Assistant) roughly every second, and immediately after any state change. Frame type byte is always 0x01 for this direction.

Status Frame Byte Layout (device -> cloud, e.g. 481101E501000000020000398C69000171)



Offset	Field	Description
0	Magic byte	Constant 0x48. Marks the start of every frame, both directions.
1	Length	Total frame length in bytes, including this byte and the final checksum.
2	Frame type	0x01 for a status report. (0x02 is used for the command direction — see Section 2.)
3	Sequence	Rolling counter incremented by the device on every frame it sends. Not used for authentication.
4	byte4	Unidentified. Observed as 0x01 in nearly every capture; role unknown.
5	byte5	Unidentified. Occasionally changes alongside light/speed during panel use; exact role unclear.
6	Light (white)	0 = off, 1 = on. The white light channel, independent of the RGB colour. Settable via Command ID 0x03.
7	Fan speed	0 = off, 1–4 = speed levels. Settable via Command ID 0x04.
8	RGB mode flag	0 = never engaged (seen only on first-ever connection). 1 = off, but colour value is remembered. 2 = on. Confirmed only ever to change via the physical panel button; no working remote command was found despite an extensive sweep (see Section 3, Command ID 0x07 investigation).
9	byte9	Unidentified. Always 0x00 in every capture so far.
10	Filter reminder flag	0 = no warning, 1 = filter needs cleaning. Confirmed 2026-08-13 against a real, organically-triggered filter warning. Mirrors byte 14 exactly — both flip together. Previously listed as unidentified/always-0x00; only ever differs from 0 once the panel's actual filter indicator is lit.
11	Red	0–255. Confirmed by cycling all 9 physical panel colour presets and cross-referencing each against this byte.
12	Green	0–255. Confirmed as above.
13	Blue	0–255. Confirmed as above.
14	Filter reminder flag	Mirrors byte 10 (see above). Reason for the duplication unknown — possibly a redundancy/checksum-adjacent design choice, possibly two related-but-distinct conditions that happen to always coincide in testing so far.

Offset	Field	Description
15	Change flag	Transient “just changed” pulse: briefly 1 right after any state change, then reverts to 0 on the next heartbeat. Ignore when reading steady-state values.
16	Checksum	Sum of every preceding byte in the frame, modulo 256 (simple byte-sum, no CRC polynomial). Confirmed identical across three independent open-source implementations of this protocol.

2. Command Frame Shapes (app/bridge → device)

Commands sent to the device follow the same magic-byte/length/type/sequence header, but with frame type 0x02, and a payload shape that depends on the Command ID. Four shapes are confirmed:

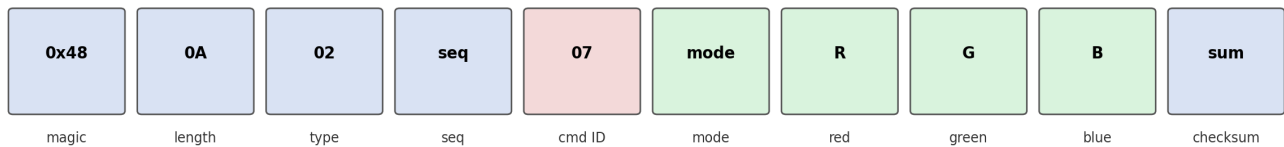
Simple command (Power 0x02, Light 0x03, Speed 0x04, Timer 0x05)

example: 480702D70301 2C -> Light ON



Colour command (cmdId 0x07 only)

example: 480A02E80702398C6973 -> set colour to #398C69, RGB on



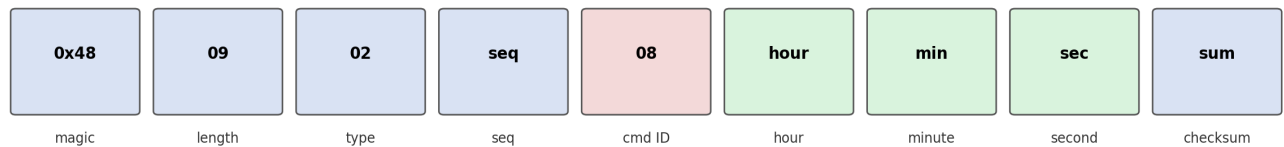
Timer command (cmdId 0x05)

example: 480702D80501 2F -> auto-off timer, duration unit 1



Clock-set command (cmdId 0x08) — confirmed 2026-08-11

example: 48090204080F270095 -> set clock to 15:39:00



Note the colour command's extra **mode** byte before R/G/B — without it, the device silently misinterprets the frame. This was the key discovery that unlocked reliable colour control.

3. Command ID Address Space

Command ID occupies byte offset 4 of a command frame and selects which function the payload targets. Every value from 0x02 through 0x14 has been tested; results below.

ID	Function	Status	Notes
0x02	Power	Working	Master fan+light power. Value byte: 0/1.
0x03	Light (white)	Working	White light only, independent of RGB. Value byte: 0/1.
0x04	Fan speed	Working	Value byte: 0–4.

ID	Function	Status	Notes
0x05	Auto-off timer	Working	Value byte sets countdown duration (units unconfirmed, roughly seconds-to-minutes). Panel-cancellable. On completion, only resets its own duration byte (offset 9) — confirmed to have zero effect on the RGB mode flag (offset 8) or any other field. A value of 0x02 produces a ~90-second countdown; retested cleanly on 2026-08-11 after an earlier ambiguous log (from 2026-07-28) raised a false alarm suggesting it might toggle the RGB flag directly — that turned out to be coincidental timing with a manual panel press, not a real effect.
0x06	Filter/clean reminder reset	Working, confirmed	Value byte 0x00. Solved 2026-08-13: earlier extensive testing (bare, 1-byte, extended 2-byte payloads) consistently found no response, which turned out to be because a filter-reset command has no observable effect while no filter warning is active. Retested against a real, organically-triggered warning and confirmed immediately — response arrived in under one second and cleared both status bytes 10 and 14, plus the physical panel icon.
0x07	RGB colour	Working	Requires 4-byte payload: mode + R + G + B. mode=0x02 sets the colour and is the only mode value the device reliably accepts. mode=0x00 and 0x03 are also accepted but behave identically to 0x02. mode=0x01 (assumed “off” by symmetry) gets zero response, every time, across dozens of tests.
0x08	Clock (time-of-day)	Working, confirmed	Two distinct uses confirmed. (1) Simple 1-byte values accidentally adjust the wall-clock display (early discovery, required a manual reset). (2) Proper 3-byte payload — [hour, minute, second], with a genuinely computed checksum, not the hardcoded 0x00 a reference implementation used — reliably sets the clock to an exact time. Confirmed working 2026-08-11 and built into the add-on as a “Sync Clock” button. The clock is never reported back in the status frame — this command is fire-and-forget, verified visually only.
0x09	RGB on/off (per community docs)	Documented, not functional	Identified via an independent open-source adapter's protocol table (cmdId 9 = “RGB_Sw”). That project's own author marked it “not implemented” after being unable to get a response. Exhaustively retested on this hood: bare, 1-byte, extended 2-byte, the reference adapter's exact literal format, a 3-byte colour-shaped payload, and the full 4-byte [mode,R,G,B] shape that unlocked cmdId 0x07 — 15 distinct variants total, tested with the RGB light confirmed genuinely on beforehand (to rule out silence-because-nothing-to-turn-off). No response in any case.
0x0A–0x0D	—	No response	Tested bare and with 1-byte values (0/1). Silent every time.
0x0E–0x14	—	No response	Tested bare, with 1-byte values, and (0x0E–0x10 only) with a 3-byte RGB-shaped payload. Silent every time.
various	Frame-type byte variants	No effect	Tested 0x00, 0x01, 0x03, 0x04 in place of the usual 0x02 frame-type byte, paired with cmdId 0x07. No response from any.

4. How This Maps to the Home Assistant Add-on

The hekr_bridge add-on builds and parses these exact frames. Practical behaviour, given everything above:

- **Power / Light / Fan Speed** entities map directly to Command IDs 0x02/0x03/0x04 — fully reliable.
- **RGB Light** entity uses Command ID 0x07. Since no working off-command exists (offset 8, mode 0x01 dead end), “off” is implemented by setting colour to black (0,0,0) via the same working mode=0x02 command. This matches Hekr’s own official MCU-SDK reference implementation, which internally implements light-off the same way.
- **Turning back on** restores the last genuine non-black colour, tracked from either Home Assistant or the physical panel — not just the last literal command sent.
- **Sync Clock** button (added 2026-08-11) uses Command ID 0x08 to set the hood’s clock to the add-on’s current system time, or an explicit override time if published to the same topic — useful if the add-on container’s timezone doesn’t match the hood’s physical location.
- **Filter Needs Cleaning** binary sensor and **Reset Filter Reminder** button (added 2026-08-13) read and clear status bytes 10/14 via Command ID 0x06 — confirmed working against a real filter warning, closing out what had been an open investigation since the project began.
- **Known limitation:** the RGB mode flag (status byte 8) only ever reaches value 1 (“off, remembers colour”) via the physical panel. The front-panel indicator light therefore stays lit when the hood is set to black via Home Assistant, even though the strip itself is dark. This remains the one genuinely unresolved item from the original investigation.

Sources cross-referenced during this investigation: direct packet capture and command sweeping against this device; Hekr’s official HEKR-MCU-SDK reference firmware; the ioBroker.hekr community adapter; the hekrapi-python protocol library; and the decompiled Wisen Android application.

Revised 2026-08-13: filter/clean reminder fully solved (indicator bytes and reset command both confirmed against a real warning).
Revised 2026-08-11: added confirmed clock-set command, retested cmdId 0x05/0x06 findings.